An Empty Region of Technical Strategy Space: 18,630 Pre-Registered Variants Across Cryptocurrency and FX

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Abstract

Published results about trading strategies pass through a filter that keeps only what worked. This paper reports the other side of that filter. Seventeen strategy families were enumerated exhaustively into 18,630 parameter variants and evaluated on hourly cryptocurrency and FX data. Every sweep was pre-registered: the search space, protocol, cost model and acceptance threshold were written down, hashed with SHA-256, and anchored to the Bitcoin blockchain before the corresponding data was loaded. The full return distribution of every variant is reported, not the best one.

Across 29 scored sweeps, none cleared a Deflated Sharpe Ratio of 0.95. The highest observed was 0.745. In 27 of the 29 sweeps the best variant performed worse than the maximum expected under the null hypothesis of no skill — that is, worse than the best of an equivalent number of random strategies. The median minimum detectable annualised Sharpe was 3.49, which bounds what this study could and could not have seen: effects below that threshold remain untested rather than refuted.

Four results follow from reporting every sweep rather than the best one. A sign test across sweeps rejects the joint null of no effect in the direction opposite to the one a reader might expect ($p = 1.6 \times 10^{-6}$): the families underperform chance collectively, not merely individually. A cost-sensitivity decomposition separates families that die under transaction costs from the seventeen of twenty-nine whose median variant is unprofitable even at zero cost. The measured kurtosis of the realised return series exceeded the value assumed at pre-registration in every single sweep, by a median factor of 354. And the two cross-sectional momentum sweeps produced positive gross Sharpe in every one of their 45 variants, with a median breakeven cost of 29 to 31 basis points against the 10 charged, while remaining inside the spread of independently shuffled panels ($z = 0.8$ and 1.8) and below the acceptance threshold — a case where two independent criteria, one analytic and one empirical, agree that an apparent effect cannot be separated from noise.

The registry also records its own failures. Three cross-sectional sweeps were voided before submission, when a weight-allocation defect was found that assigned the entire short leg of half their variants to assets outside the investable universe. They were re-registered, re-timestamped and re-run; both the voided records and their replacements are reported here. One of the three replacements is the exact arithmetic inverse of another and is marked as one piece of evidence rather than two.

The contribution is not a claim that technical trading does not work. It is a mapped region of strategy space with a stated detection limit, published whether or not it contained anything, together with the timestamps that let a third party verify the pre-registration independently of the author. Every sweep, including the eight that were voided and the reasons for voiding them, is reported in the appendix.

1 Introduction

Consider a thousand researchers independently testing the same worthless strategy. Roughly fifty will observe a result significant at the five percent level. If only those fifty publish, the

literature will contain fifty confirmations and no refutations, and a reader cannot distinguish that situation from one in which the strategy works.

This is not a hypothetical concern in quantitative finance. Backtests are cheap, the space of testable rules is effectively unbounded, and the incentive to report only the surviving variant is strong. Bailey and López de Prado (2014) showed that with enough trials a backtest with an in-sample Sharpe ratio of 1.0 can be produced from pure noise, and derived the Deflated Sharpe Ratio to correct for it. Harvey et al. (2016) argued that most published cross-sectional return predictors fail once multiple testing is accounted for. López de Prado (2018) described backtest overfitting as the central methodological problem of the field.

The corrections exist. What is missing is the negative record: systematic publication of regions that were searched and found empty. Without it, the corrections are applied to a sample that has already been filtered.

This paper reports one such region. It is small relative to the space of possible strategies, and the result will not surprise anyone who has internalised the multiple-testing literature. Its value is that the search space, the protocol and the threshold were fixed and cryptographically timestamped before the data was examined, so the absence of a positive result cannot be attributed to a threshold chosen after the fact.

2 Method

2.1 The unit of publication

The unit here is not a strategy but a *region of strategy space*. The statement "RSI(14) did not work on BTC" carries almost no information: it describes one point in a continuum, and the reader cannot tell whether a neighbouring parameter would have worked. The statement "the entire short-horizon mean reversion family, 1,620 enumerated variants, across two symbols over eight years, produced nothing above a stated detection limit" is a claim someone else can build on or refute.

Every sweep therefore enumerates a complete parameter grid rather than selecting parameters. No variant is favoured, and the whole distribution is reported.

2.2 Pre-registration and timestamping

For each sweep a JSON document was written specifying the strategy family and its full parameter grid; the symbols, timeframe and date range; the execution and cost model; the planning assumptions used for the power calculation; and the acceptance threshold. The document was canonicalised, hashed with SHA-256, and the hash submitted to OpenTimestamps (Todd, 2016), which anchors it in a Bitcoin transaction.

The ordering this establishes is the point. A reader can verify that the document existed in its current form before the block containing its hash was mined, and therefore before any result reported here. Only the hash is published to the timestamping service; the document itself stays with the author until release.

All 38 pre-registrations in the registry carry such a timestamp.

Records are additionally linked in a hash chain,

$$\text{chain}_i = \text{sha256}(\text{chain}_{i-1} \,\|\, \text{id}_i \,\|\, \text{sha256}(\text{prereg}_i)) \,,$$

with $\text{chain}_0 = \text{sha256}(\texttt{"sweep-registry-genesis"})$. Timestamping alone proves that each document predates its result but says nothing about deletion: an inconvenient record could be removed quietly and every remaining one would still verify. Under the chain, removing a record changes every subsequent link, including the published head. For a registry whose contents are almost entirely negative results, that is the failure mode worth defending against.

2.3 Implementation

Each variant maps a signal to a target weight at the close of bar t, held over bar $t+1$. Signals are computed only from information available at or before the close of t; no indicator uses a centred window, and no rebalance uses the bar it is executed in. Within a sweep every variant sees the same bars, so cross-variant comparisons are not contaminated by differing sample lengths.

Costs are charged on the absolute change in weight,

$$r_t^{\text{net}} = w_{t-1} r_t - c\,|w_{t-1} - w_{t-2}|,$$

where c is the round-trip cost in basis points fixed at pre-registration and never adjusted after seeing a result: 4 bps commission plus 6 bps slippage for the cross-sectional sweeps, which trade altcoins outside the majors and for SW-0036; 4 plus 2 for the other single-asset crypto sweeps; and 1 plus 0.5 for the FX sweeps, where major-pair spreads are a fraction of those in crypto. The lower FX charge means a smaller true edge would have survived there, so a null result on FX is stronger evidence than a null on crypto, not weaker.

Sharpe ratios are annualised as $\sqrt{P}\,\bar{r}/s_r$ with $P = 8{,}760$, the number of periods per year declared in every pre-registration. For the FX sweeps this treats the five-day trading week as if it were continuous, which inflates the annualisation factor by roughly 18%; the same constant is applied to every variant within a sweep, so cross-variant comparisons and the sign of each result are unaffected, but the FX Sharpe levels reported here are not directly comparable to figures annualised on a 252-day convention. No risk-free rate is subtracted; over this sample the adjustment is small relative to the dispersion across variants, and omitting it is conservative in the sense that it inflates rather than deflates the reported figures.

For the cross-sectional sweeps the investable universe is rebuilt monthly from the *previous* month's quote volume. Weights are assigned by rank within the universe, demeaned to be dollar-neutral, and normalised to unit gross exposure.

The returns matrix for each sweep is stored as a variant-by-observation array of 32-bit floats and memory-mapped at scoring time; per-variant Sharpe ratios are accumulated in column chunks. This matters only because it makes the raw matrices, rather than summary statistics, publishable alongside each record.

2.4 Evaluation

Scoring uses the Deflated Sharpe Ratio of Bailey and López de Prado (2014), which adjusts the observed maximum Sharpe for the number of trials, the non-normality of the return distribution and the sample length. Alongside it we report:

- $E[\max SR]$ under the null — the best Sharpe one expects from N independent worthless strategies. A best variant below this figure is evidence *against* the family, not merely absence of evidence for it.
- The minimum detectable Sharpe — the smallest true effect the design had 80% power to distinguish at $\alpha = 0.05$. This bounds the claim.
- The Probability of Backtest Overfitting (PBO) via combinatorially symmetric cross-validation (Bailey et al., 2017), over 12,870 combinations from 16 splits, for every sweep.
- The full quantile distribution across variants, and the share of variants with positive Sharpe.

Skewness and kurtosis are measured from the realised matrix rather than assumed; the planning assumptions are recorded separately in each pre-registration so that the two can be compared.

2.5 Survivorship

Ammann et al. (2022) measured an annualised survivorship bias of 62.19% for equal-weighted cryptocurrency portfolios constructed from a present-day universe. Point-in-time construction means a symbol is held only in months when it would actually have been selected, and symbols that later delisted are retained for the period they traded. Without it, almost any result in this asset class is uninterpretable.

3 Scope

Pre-registered sweeps in the registry	38
Bitcoin-timestamped	38
Scored	29
Voided (see §5)	8
Sealed, window not yet open	1
Variants enumerated across scored sweeps	18,630
of which produced at least one trade	17,811
Strategy families with a scored sweep	17

Table 1: Registry scope, October 2026.

Data are hourly bars from Binance public archives (spot, USDT pairs) for BTCUSDT and ETHUSDT, and hourly bars for EURUSD, GBPUSD and USDJPY. The single-asset sweeps span 2017-09-01 to 2025-06-30; the cross-sectional sweeps span 2019-02-01 to 2025-06-30, limited by the point-in-time universe construction. Everything from 2025-07-01 onward is reserved as an untouched holdout. Cryptocurrency accounts for 17 scored sweeps and 9,153 variants, of which three sweeps and 135 variants are cross-sectional; FX for 12 sweeps and 9,477 variants.

The realised observation counts differ marginally from those assumed at pre-registration: 48,749 against 48,772 planned for the FX sweeps (23 bars, from gaps in the source archive) and 56,870 against 56,000 planned for SW-0036. Both are recorded in the respective results.

The seventeen families are: short-horizon mean reversion; moving-average crossover; Donchian channel breakout; ATR volatility breakout; time-series momentum; volatility-filtered momentum; hour-of-day seasonality; volume-confirmed price signals; six price-action patterns commonly grouped as "smart money concepts" (fair value gap, order block, liquidity sweep, break of structure, engulfing candle, inside bar breakout); and three cross-sectional families (momentum, reversal, volatility-scaled momentum). Twelve of the seventeen were run on both crypto and FX.

4 Results

4.1 Headline

No sweep cleared the pre-specified Deflated Sharpe Ratio threshold of 0.95. The maximum observed was 0.745 (volatility-scaled cross-sectional momentum, SW-0041), followed by 0.605 (cross-sectional momentum, SW-0039). Every other family scored below 0.5, and ten of the seventeen scored below 0.01.

The two leading figures come from the re-run replacements described in §5, not from the voided originals, whose higher values (0.823 and 0.809) were produced by the defect reported there.

Family	Sweeps	Variants	Best DSR
Vol-scaled cross-sectional momentum	1	45	0.745
Cross-sectional momentum	1	45	0.605
Break of structure	2	1,350	0.482
Donchian channel breakout	2	1,080	0.415
ATR volatility breakout	2	1,350	0.255
Moving-average crossover	2	1,575	0.225
Volatility-filtered momentum	2	900	0.010
Inside bar breakout	2	1,350	0.002
Cross-sectional reversal	1	45	0.001
Time-series momentum	2	1,350	0.0003
Liquidity sweep	2	1,350	0.00006
Short-horizon mean reversion	1	1,620	0.000
Hour-of-day seasonality	2	1,440	0.000
Fair value gap	2	1,350	0.000
Order block	2	1,350	0.000
Engulfing candle	2	1,350	0.000
Volume-confirmed price signals	1	1,080	0.000

Table 2: Best Deflated Sharpe Ratio by family across scored sweeps. Pre-specified threshold 0.95. Variant counts are the enumerated grids.

4.2 Worse than chance

In 27 of the 29 scored sweeps the best variant found was *below* $E[\max SR]$ under the null. This is a stronger statement than failing to reach significance. It says that enumerating the family and taking its best member produced less than enumerating an equal number of random strategies and taking their best member would have.

Figure 1 shows why. For most sweeps the entire interquartile range sits below zero, and the blue marker — chance expectation — lies to the right of the best variant found. The two cross-sectional momentum sweeps at the top of the figure are the visible exceptions: their distributions sit almost entirely above zero.

The obvious explanation is transaction cost, and for some families that is correct. But the diagnostics reported in §4.4 show it is the minority case: for seventeen of the twenty-nine sweeps with a cost decomposition, the median variant is unprofitable *before* any cost is charged. These rules do not merely fail to overcome friction. They carry predictive content pointing the wrong way.

4.3 A joint test

Twenty-nine separate tests, none of which rejects, is a weak statement. The sweeps can instead be treated as one sample. Under the null that no family carries an effect, the best variant of a sweep should fall above or below $E[\max SR]$ with roughly equal probability, since $E[\max SR]$ approximates the centre of the distribution of the maximum. Observing 27 of 29 on one side has a two-sided binomial probability of 1.6×10^{-6}.

The direction matters. This is not a rejection in favour of the strategies; it is a rejection against them. The median ratio of best variant to chance expectation is 0.321, meaning the typical family's best member reached only a third of what an equal number of coin flips would have produced.

The joint test is worth stating precisely because it sidesteps the detection limit that weakens each sweep individually. No single sweep had the power to distinguish a modest effect from zero. The collection has ample power to distinguish "these families are slightly better than chance" from "these families are substantially worse than chance", and it selects the latter.

Two caveats. The approximation $P(\text{best} < E[\max SR]) = 0.5$ under the null is exact only for a symmetric distribution of the maximum, which is not guaranteed; the true null probability is somewhat above 0.5, which makes the test conservative in the direction reported. And the sweeps are not fully independent: twelve families were run on both markets, and the crypto and FX samples overlap in time. The effective sample is therefore smaller than 29, though not by enough to overturn a p-value of this magnitude.

4.4 Cost sensitivity

Because gross and net returns were both recorded, the cost assumption can be examined without re-running anything. Table 8 reports, for each sweep, the median variant's annualised Sharpe gross of cost and net at the cost actually charged, together with the breakeven cost.

Only measured points are tabulated. An earlier draft reported a grid of 0, 2, 6, 10 and 20 basis points, obtained by assuming Sharpe falls linearly in the per-unit cost and interpolating from the two measured values. That assumption is wrong. Mean return is linear in cost, but the standard deviation of the net series is not, because per-bar cost scales with per-bar turnover, which varies across the sample; for SW-0041 the 5th and 95th percentiles of turnover per bar are 0.012 and 0.112, roughly a ninefold spread. Only the two endpoints were ever computed from returns, so the three intermediate columns were published numbers that had not been measured. They are removed.

Breakeven is defined as the per-variant round-trip cost at which that variant's mean gross return reaches zero; the figure reported is the median *of those breakeven points*, which is not in general the breakeven of the median variant. Earlier text described it as the latter.

The decomposition splits the registry cleanly:

- **Twelve sweeps have a positive median gross Sharpe**, and eight of the twelve are killed by friction. Donchian breakout and moving-average crossover on crypto are the only two single-asset sweeps whose breakeven cost (8.6 and 7.9 bps) exceeds the 6 bps charged, and whose median variant therefore remains positive after costs. The two cross-sectional momentum sweeps clear their higher charge by a wider margin (29.0 and 30.7 bps against 10), which is the subject of §4.8.

- **Seventeen sweeps have a negative median gross Sharpe.** No cost assumption rescues them. Inside bar breakout on crypto is the extreme case: a breakeven cost of −15.2 bps, meaning the median variant loses money even if the venue paid it 15 bps per round trip to trade.

The median breakeven across all twenty-nine is −0.03 bps. This is the single most useful number for a practitioner reading this paper, and it is only available because the gross series was retained.

The cost decomposition now covers all 29 scored sweeps. The three cross-sectional records that previously lacked one were voided and re-run as SW-0039, SW-0040 and SW-0041 with diagnostics enabled; their breakeven figures, 29.00 and 30.68 bps for the two momentum sweeps, are the highest in the registry and are discussed in §4.8.

4.5 Calibration of the power analysis

Every pre-registration assumed a kurtosis of 8.0 and a skewness of −0.2 when computing its detection limit. These were not arbitrary: 8.0 is a reasonable figure for hourly returns on a liquid asset, and the planning assumptions were frozen before any data was loaded, which is the point.

They were wrong in every sweep, in the same direction, by a large margin.

	Assumed	Measured (median)
Kurtosis	8.0	2,835.3
Skewness	−0.2	28.9
Sweeps where measured kurtosis exceeded assumed		29 of 29
Median underestimation factor		354×
Maximum underestimation factor		2,666×

Table 3: Planning assumptions against realised moments.

The gap is not a defect in the assumption so much as a property of strategy returns that is easy to overlook. The assumed figure describes the *asset*; the realised figure describes the *strategy's return series*, which is the asset return multiplied by a position that is frequently zero and occasionally large. The skewness figure moves in the opposite direction to the assumption rather than merely exceeding it: −0.2 was assumed, and the median realised value is +28.9, positive in 27 of 29 sweeps. The two quantities move together — the sweeps with the largest skewness are the sweeps with the largest kurtosis (SW-0010 at 128.5 and 17,961; SW-0027 at 123.3 and 19,466) — which is what a return series of mostly zeros punctuated by rare large moves produces. The sign of the assumption was taken from asset returns, where negative skewness is typical; it does not carry over to the conditioned series.

Conditioning on a trading rule concentrates the distribution's mass at zero and leaves the tails intact, and kurtosis responds to exactly that shape.

The practical implication is that power analyses for strategy sweeps that plug in asset-level moments will understate the detection limit, in this sample by a median factor of 354 in kurtosis and correspondingly in the derived threshold. This is a correctable mistake, and it is correctable only if someone reports both numbers.

4.6 What actually drives the Deflated Sharpe Ratio

The previous result has a consequence that is uncomfortable for the metric this paper uses as its primary criterion.

Spearman correlation	ρ
Measured kurtosis vs. DSR	−0.728
Best variant Sharpe vs. DSR	+0.491
Measured kurtosis vs. $E[\max SR]$	+0.565
Mean exposure vs. measured kurtosis	−0.194

Table 4: Rank correlations across the 29 scored sweeps.

Across these sweeps, the Deflated Sharpe Ratio is more strongly associated with the kurtosis of the return series than with how well the best strategy performed. The path is visible in the third row: high kurtosis inflates $E[\max SR]$, and an inflated null threshold drives the DSR toward zero regardless of what the best variant achieved.

Splitting at a kurtosis of 200 makes the size of the effect concrete:

	n	Median DSR	Median $E[\max SR]$	Median best
Kurtosis < 200	8	0.122	1.70	1.037
Kurtosis ≥ 200	21	0.000	2.53	0.779

Table 5: The DSR gap between the two groups is far larger than the gap in underlying performance.

The best variants of the two groups differ by 0.26 in annualised Sharpe. Their chance thresholds differ by half again, and their Deflated Sharpe Ratios differ by the entire available range. A reader comparing two published DSR figures is, on this evidence, comparing the tail behaviour of two return series at least as much as the quality of two strategies.

This should not be read as a claim that the Deflated Sharpe Ratio is wrong. The non-normality adjustment it applies is correct, and a strategy whose returns are a long series of zeros punctuated by rare large moves genuinely does have a noisier maximum than one with well-behaved returns. The point is narrower: the magnitude of that adjustment in realistic strategy data is large enough to dominate the comparison, and a DSR reported without the accompanying kurtosis is not interpretable.

Three limits on this finding. It is a correlation over 29 observations. Strategy families differ simultaneously in kurtosis and in quality, and this sample cannot separate the two. And an obvious candidate mechanism — that sparse, infrequently-held positions produce both the high kurtosis and the poor performance — is not supported: mean exposure and kurtosis are essentially uncorrelated ($\rho = -0.19$).

4.7 Overfitting probability

Reporting PBO for every sweep rather than only the best one produces a pattern worth stating explicitly. The median PBO across the 29 sweeps is 0.188, which is low. Taken at face value this says the variant ranking is usually stable out of sample.

That is true, and it is not good news for the strategies. When a family is reliably unprofitable, the ranking within it is stable precisely because the ordering is driven by a real and persistent quantity — turnover cost — rather than by noise. Hour-of-day seasonality scores a PBO of exactly 0.000 on both markets: the worst variants stay worst. Low PBO is informative about whether a selection procedure generalises, not about whether the thing selected is worth holding.

The highest PBO figures appear where they should: liquidity sweep on FX at 0.573, volatility-scaled cross-sectional momentum at 0.399, cross-sectional momentum at 0.339. These are the sweeps whose top variants are closest to being interesting, and they are also the sweeps where selecting the top variant is closest to a coin flip.

4.8 The best result in the registry

The strongest sweep, SW-0041, tests volatility-scaled cross-sectional momentum on a point-in-time top-20 crypto universe: assets ranked by trailing return divided by trailing volatility rather than by raw return, 45 fully enumerated variants over 56,126 hourly observations. It supersedes SW-0034, voided as described in §5.

This sweep is the one case in the registry where the surface statistics look like an effect, and it is worth setting out exactly what they show and what they do not.

Every one of the 45 variants has positive gross Sharpe, with a mean of 0.823. The portfolio is genuinely dollar-neutral — mean directional tilt is zero to four decimal places and mean market beta is +0.027 against an equal-weighted index of the investable universe — so this is not the market trend reappearing in another form, which is the failure mode that contaminated the voided predecessor. The market-neutral residual Sharpe, 0.812, is almost identical to the raw gross figure. The median variant breaks even at 30.68 basis points of round-trip cost against the 10 charged, so the effect is not one that only exists at zero cost. Cross-sectional momentum (SW-0039) shows the same pattern at slightly lower magnitude: 100% positive gross Sharpe, median breakeven 29.00 bps, mean beta −0.096.

Two independent criteria nevertheless place this inside the noise.

The first is the pre-registered one. The best net Sharpe, 1.043, falls short of the 1.453 required to clear a Deflated Sharpe Ratio of 0.95 given 45 trials and the realised moments. The

Best annualised Sharpe, net	1.0431
$E[\max SR]$ under the null	0.7744
Critical Sharpe at the threshold	1.4528
Deflated Sharpe Ratio	0.745
Probability of Backtest Overfitting	0.399
Minimum detectable Sharpe (realised)	1.804
Gross Sharpe: mean / median / max	0.823 / 0.814 / 1.529
Share of variants with positive gross Sharpe	100%
Share with positive net Sharpe	86.7%
Median breakeven cost	30.68 bps
Cost charged	10.0 bps
Mean gross exposure	1.988
Mean directional tilt	−0.0000
Mean market beta	+0.027
Market-neutral residual Sharpe, mean	0.812
Residual Sharpe on shuffled panels	-0.079 ± 0.491
Observed residual Sharpe	+0.812
z against the shuffled null	1.816

Table 6: SW-0041 in full. Verdict: no edge found.

PBO of 0.399 says the same thing from a different direction: the variant that ranks best in a training split lands below the out-of-sample median in 40% of the 12,870 splits.

The second was computed after the fact and is reported as exploratory. Each asset's returns were shuffled independently, destroying both serial and cross-sectional structure while preserving every per-asset return distribution, and the whole sweep re-run on ten such panels. The mean residual Sharpe across those panels was −0.079 with a standard deviation of 0.491. The observed +0.812 sits 1.82 standard deviations above that, which is inside the spread. For SW-0039 the equivalent figure is 0.81. The instrument cannot separate this from a panel with no structure in it.

The gap between the two framings is the useful part. A reader shown only the gross Sharpe distribution — all 45 variants positive, breakeven at three times the cost charged, beta of 0.03 — would reasonably conclude that something is there. A reader shown only the DSR would conclude nothing is. Both were computed from the same returns. The pre-registered rule decides, and it says no; the shuffled-panel calibration, which was not pre-registered, happens to agree. This paper reports the disagreement rather than the figure that reads better, which is the only reason the comparison is available at all.

A holdout sweep over the same parameter space (SW-0035) was pre-registered and timestamped before its window opened. It remains unrun, and its window does not close until 2029. Running it early at reduced power would consume a single-use test to obtain a result the power analysis already predicts would be inconclusive.

4.9 Detection limit

The median minimum detectable annualised Sharpe across scored sweeps was 3.49. This is high, and it is the most important caveat in the paper. A real effect with a Sharpe of 0.8 — which would be a good strategy — sits well below this threshold for most sweeps and would not have been detected.

The limit varies enormously across sweeps, from 1.80 for the cross-sectional sweeps to 29.41 for hour-of-day seasonality on FX. The driver is the dispersion of variant Sharpe ratios within the family: a family whose variants range from −17.6 to +0.8 carries so much internal variance that no realistic effect could be distinguished from it. Those sweeps are, in practice, uninformative,

and the appendix reports their detection limits so that a reader can discount them rather than count them as evidence.

The correct reading of the headline result is therefore: *within this region, no effect large enough to clear the stated detection limit exists.* Smaller effects are untested, not refuted.

5 Corrections and integrity notes

5.1 Six voided runs, in two waves

Eight records in the registry are marked void. Six are the failed executions below; the other two are the declined runs of §5.2, voided for a different reason.

First wave: an empty matrix scored as a null result. Three sweeps (SW-0029, SW-0030, SW-0031) produced returns matrices that were identically zero. The sweep process had been killed by the operating system's out-of-memory handler partway through, leaving the allocated matrix unwritten. The scoring pipeline then accepted the empty matrix as a valid null result and reported "best Sharpe 0.0, zero variants positive", with a Deflated Sharpe Ratio of exactly 0.5. It was not a finding; it was a crashed run that resembled one, which is the worst output this registry could produce. A guard now refuses to score a matrix in which every variant has a Sharpe of exactly zero. The three hypotheses were re-run as SW-0032, SW-0033 and SW-0034.

Second wave: a weight-allocation defect in the cross-sectional engine. Those replacements were themselves voided. An earlier draft of this paper reported SW-0034 and SW-0032 as the two sweeps whose best variant exceeded chance expectation, with Deflated Sharpe Ratios of 0.823 and 0.809. A line-by-line audit of the execution engine, undertaken before submission, found the following in the cross-sectional weight allocator: assets outside the point-in-time investable universe were assigned a score of $-\infty$ and therefore sorted to the *front* of the ranking, so the short leg — taken from the front — was allocated entirely to non-investable assets whose returns are zeroed by construction. Half the variants in each sweep, the 45 long/short ones out of 90, held no short position at all. They were long-only books at gross exposure 2.0 wearing the turnover cost of a short leg that did not exist.

The diagnosis was confirmed rather than inferred. The stored returns matrices reproduce to within 2.6×10^{-8} under a replay of the defective allocator and diverge materially under the corrected one, with per-variant correlations of 0.54 to 0.69 between the two. The long-only branch was untouched by the defect, but corrupted variants still enter the trial count, the expected-maximum-Sharpe null and the CSCV combinatorics, so the primary metric of all three sweeps was invalid.

The three were voided, the defect fixed, and a regression test added covering universe containment, dollar neutrality, gross exposure and leg overlap. The hypotheses were re-registered as SW-0039, SW-0040 and SW-0041, re-hashed, re-anchored to the Bitcoin blockchain, and re-run. One deliberate change was made to the search space and recorded in the new pre-registrations: the long-only exit rule was removed, because a long-only book at gross exposure 2.0 does not test a cross-sectional hypothesis. That exclusion was decided after seeing which branch had dominated the voided results and is marked as such; it halves the variant count, which lowers the multiple-testing penalty and makes the remaining test easier rather than harder to pass. The replacements scored 0.605, 0.001 and 0.745.

The pre-registrations and Bitcoin timestamps of all six voided records remain valid; only their execution failed. Both the voided records and their replacements are public, with the reason and the superseding identifier on each page.

A record that is not independent evidence. SW-0040 (cross-sectional reversal) is implemented as the negation of the momentum score, so with a symmetric long/short fraction its portfolio is the arithmetic inverse of SW-0039. Measured per-variant correlation between the two returns matrices is -0.997 to -0.999, and their gross Sharpe, residual Sharpe, turnover and breakeven figures are mirror images to four decimal places. Only net Sharpe differs non-trivially, because transaction costs subtract from both legs rather than reflecting. The two records are one piece of evidence with a sign, not two, and both carry an annotation saying so. They are counted separately in the sweep totals, which overstates the independent evidence in the cross-sectional family by one sweep.

5.2 Two declined runs

SW-0003 tested short-horizon continuation on BTC and ETH as a follow-on to SW-0002. Its discovery sample produced a best annualised Sharpe of 1.489 against $E[\max SR]$ of 2.068 and a DSR of 0.050. SW-0004, the holdout reserved for it, was voided as declined rather than failed: a confirmatory test on a hypothesis the discovery sample has already refuted adds no information, and running it would only consume a sealed window. SW-0003 is excluded from the 29 scored sweeps and from all totals reported here.

5.3 Symbols counted as independent trials

In the 26 multi-symbol sweeps — 17,676 of the 17,811 scored variants, or 99% of the registry — a variant is one parameter combination applied to one symbol, so the same combination run on BTCUSDT and on ETHUSDT counts as two trials. SW-0008 is typical: 270 parameter combinations across two symbols are enumerated and scored as 540 trials.

This affects the primary metric in two ways, both in the same direction. The Deflated Sharpe Ratio penalises by trial count, so scoring 540 rather than 270 applies a heavier multiple-testing penalty than the number of distinct hypotheses warrants. And $E[\max SR]$ assumes the trials are independent, which paired symbols are not: the same rule on BTCUSDT and ETHUSDT produces correlated return series, so the effective number of independent trials is below the nominal count, and the null threshold is correspondingly higher than an effective-N calculation would give.

Both errors therefore run in the conservative direction: they make the threshold harder to clear, not easier. This is the opposite of the bias described in the next subsection, and it is reported for the same reason — a reader recomputing these figures with distinct parameter combinations as the unit, or with an effective trial count, would obtain Deflated Sharpe Ratios somewhat higher than those reported here. No sweep is close enough to the 0.95 threshold for this to change a verdict: the largest, SW-0041, is a single-universe cross-sectional sweep with no symbol multiplication at all.

5.4 Variants that never traded

In seven of the 29 scored sweeps the number of variants scored is smaller than the number enumerated, 17,811 against 18,630 in total. The difference consists entirely of variants whose parameter combination produced no trade at all over the sample: the Sharpe of an identically zero return series is undefined, and such columns were excluded at scoring time.

The scoring pipeline flags this automatically. Every affected record carries an integrity note of the form *"Trial count differs from pre-registration: declared 540, supplied 261. Explain this in the record."*, and the explanation is recorded in that sweep's `annotations.json`.

The exclusion is not neutral. The Deflated Sharpe Ratio penalises by trial count, so scoring 261 trials rather than the 540 enumerated applies a lighter multiple-testing penalty and biases

the DSR *upward*. The error, in other words, runs in the permissive direction. It is reported here for that reason rather than despite it.

Its practical effect on the conclusions is nil. The largest DSR among the seven affected sweeps is 0.002, and the two with the largest shortfall (SW-0022 and SW-0028, inside bar breakout, 261 of 540 and 405 of 810) scored 0.002 and 0.00007. The two sweeps carrying the paper's highest scores, SW-0039 and SW-0041, had no excluded variants: all 45 traded in each.

These corrections are reported because a registry that conceals its own failed runs reproduces the problem it exists to address.

6 What this study does not show

1. **That technical trading does not work.** The tested region is small relative to the space of possible strategies. Nothing here addresses discretionary trading, execution-based strategies, alternative data, or any family not enumerated.

2. **That effects below the detection limit are absent.** With a median minimum detectable Sharpe of 3.49, this study is uninformative about modest but real edges.

3. **That the cost model is correct.** Costs were fixed in advance, which prevents tuning, but a wrong fixed assumption is still wrong. The direction matters: if real costs are lower than assumed, some families would score better.

4. **Generality across timeframes.** All sweeps use hourly bars. Higher-frequency effects are outside scope entirely.

5. **Generality across venues or instruments.** Cryptocurrency data comes from a single exchange archive, and the single-asset sweeps use two crypto symbols and three FX pairs. Cross-venue differences are not examined.

6. **Freedom from survivorship bias in every sweep.** Point-in-time universe construction was applied to the cross-sectional sweeps. The fixed-symbol sweeps use symbols that still trade today, which for BTC, ETH and the three FX majors is a mild assumption but not a vacuous one.

7 Reproduction

The registry — every pre-registration, its canonical frozen form, its SHA-256 hash, its OpenTimestamps proof, the scoring output, the diagnostics and the annotations — is at https://github.com/mobinmoharami/sweep-registry. Data are downloadable from Binance public archives with the included scripts; no API key or account is required.

The returns matrices themselves are not in the repository. They total 4.5 GB, which exceeds both the per-file and per-repository limits of the hosting service; the largest single matrix is 444 MB. What is published instead is `RETURNS_CHECKSUMS.json`, giving the SHA-256 of every matrix, together with the scripts that regenerate them. A third party who re-runs a sweep can confirm bit-for-bit that the matrix scored here is the matrix their run produced:

```
python run_sweep.py sweeps/SW-0008
sha256sum sweeps/SW-0008/returns.npy    # compare to RETURNS_CHECKSUMS.json
python finalize.py sweeps/SW-0008
```

This is weaker than publishing the matrices directly, and the author intends to deposit them with a data repository that accepts archives of this size.

Every number, table and figure in this paper is regenerated from the registry by `paper/analysis.py`, which reads the sweep directories and hard-codes nothing. Running it writes `paper/build/stats.txt`, in which each figure quoted here appears with its label; any discrepancy between that file and this text is a defect in the text.

To verify that a pre-registration predates its result, recompute the SHA-256 of the frozen document and check it against the `.ots` proof with the OpenTimestamps client. The proof resolves to a Bitcoin block height and timestamp. The chain view recomputes every hash from the frozen documents on request, so a record altered after sealing reports as broken rather than continuing to display a stale verification badge.

8 Conclusion

Seventeen strategy families, enumerated into 18,630 variants across two asset classes and up to eight years of hourly data, produced nothing above a Deflated Sharpe Ratio of 0.745 against a pre-specified threshold of 0.95. In 27 of 29 scored sweeps the best variant performed worse than chance expectation, and the single strongest result carries a Probability of Backtest Overfitting of 0.40.

Three findings survive independently of whether anyone cares about these particular families. A sign test across sweeps rejects the joint null against the strategies at $p = 1.6 \times 10^{-6}$, which no individual sweep had the power to do. A cost decomposition shows that most of these families are unprofitable before friction rather than because of it. And the moments assumed in every power analysis were wrong in the same direction in every sweep, by a median factor of 354, with the consequence that the Deflated Sharpe Ratio in this sample tracks kurtosis more closely than performance.

The methodological claim is narrower and more durable than the empirical one: pre-registration with cryptographic timestamping is cheap, it is practical for an individual researcher rather than only for institutions, and publishing the full distribution of a negative sweep costs nothing while being worth more than another positive result of unknown provenance.

Disclosure

The author used an AI assistant for drafting and editing this manuscript. All data collection, code, analysis and conclusions are the author's own, and the author takes full responsibility for the content.

Data and code availability

All materials are public at https://github.com/mobinmoharami/sweep-registry under the terms stated in that repository.

A All scored sweeps

Table 7: Every scored sweep. Declared is the enumerated grid; scored excludes variants that never traded. Best and $E[\max SR]$ are annualised Sharpe ratios; MDS is the minimum detectable annualised Sharpe at 80% power; PBO is the probability of backtest overfitting.

ID	Family	Market	Decl.	Scored	Best	$E[\max]$	DSR	MDS	PBO
SW-0002	Mean reversion	BTC/ETH	1,620	1,599	0.640	2.533	0.000	4.01	0.198
SW-0005	MA crossover	BTC/ETH	630	630	0.958	1.228	0.225	2.12	0.175

ID	Family	Market	Decl.	Scored	Best	E[max]	DSR	MDS	PBO
SW-0006	Donchian breakout	BTC/ETH	432	432	1.134	1.211	0.415	2.10	0.124
SW-0007	ATR breakout	BTC/ETH	540	540	1.262	1.493	0.255	2.36	0.035
SW-0008	TS momentum	BTC/ETH	540	540	1.112	2.335	0.000	3.23	0.079
SW-0009	Vol-filtered mom.	BTC/ETH	360	360	0.962	1.797	0.010	2.68	0.206
SW-0010	Hour-of-day	BTC/ETH	576	552	0.646	5.755	0.000	11.08	0.000
SW-0011	MA crossover	FX	945	945	0.810	1.688	0.019	2.73	0.354
SW-0012	Donchian breakout	FX	648	648	1.092	2.330	0.000	3.20	0.243
SW-0013	ATR breakout	FX	810	810	1.095	2.822	0.000	3.98	0.128
SW-0014	TS momentum	FX	810	792	1.102	3.774	0.000	4.90	0.261
SW-0015	Vol-filtered mom.	FX	540	540	0.880	3.414	0.000	4.50	0.122
SW-0016	Hour-of-day	FX	864	828	0.787	13.990	0.000	29.41	0.000
SW-0017	Fair value gap	BTC/ETH	540	540	0.543	2.273	0.000	3.39	0.256
SW-0018	Order block	BTC/ETH	540	504	0.349	3.498	0.000	7.12	0.098
SW-0019	Liquidity sweep	BTC/ETH	540	540	0.635	1.681	0.000	2.71	0.188
SW-0020	Break of structure	BTC/ETH	540	540	1.693	1.709	0.482	2.59	0.011
SW-0021	Engulfing candle	BTC/ETH	540	540	0.553	4.442	0.000	9.57	0.007
SW-0022	Inside bar breakout	BTC/ETH	540	261	0.410	1.440	0.002	2.74	0.302
SW-0023	Fair value gap	FX	810	810	0.650	4.633	0.000	8.60	0.035
SW-0024	Order block	FX	810	810	0.874	7.465	0.000	12.79	0.003
SW-0025	Liquidity sweep	FX	810	810	0.853	2.505	0.000	3.98	0.573
SW-0026	Break of structure	FX	810	810	0.800	2.491	0.000	3.94	0.220
SW-0027	Engulfing candle	FX	810	810	0.684	8.387	0.000	21.63	0.000
SW-0028	Inside bar breakout	FX	810	405	0.779	2.396	0.000	3.49	0.348
SW-0036	Volume-confirmed	BTC/ETH	1,080	1,080	0.864	7.920	0.000	8.93	0.221
SW-0039	XS momentum	XS crypto	45	45	0.970	0.859	0.605	1.95	0.339
SW-0040	XS reversal	XS crypto	45	45	-0.203	1.012	0.001	1.94	0.348
SW-0041	XS vol-scaled mom.	XS crypto	45	45	1.043	0.774	0.745	1.80	0.399

B Cost sensitivity

Table 8: Median variant annualised Sharpe, gross and net at the cost charged, by sweep. Both columns are measured; no values are interpolated. Breakeven is the median of the per-variant round-trip costs at which mean gross return reaches zero; negative means no cost level makes that variant profitable. All 29 scored sweeps carry diagnostics.

ID	Gross	Cost (bps)	Net	Breakeven
SW-0002	-0.209	6.0	-0.554	-4.80
SW-0005	0.336	6.0	0.057	7.91
SW-0006	0.335	6.0	0.103	8.65
SW-0007	0.214	6.0	-0.075	3.82
SW-0008	0.214	6.0	-0.127	3.33
SW-0009	0.298	6.0	-0.128	3.75
SW-0010	0.078	6.0	-1.462	0.29
SW-0011	-0.040	1.5	-0.541	-0.09
SW-0012	-0.060	1.5	-0.559	-0.12
SW-0013	0.163	1.5	-0.562	0.32
SW-0014	-0.051	1.5	-0.359	-0.16
SW-0015	-0.033	1.5	-1.024	-0.03
SW-0016	-0.025	1.5	-2.850	-0.01
SW-0017	-0.034	6.0	-0.631	-0.92
SW-0018	-0.149	6.0	-1.071	-0.85
SW-0019	-0.125	6.0	-0.348	-4.42

ID	Gross	Cost (bps)	Net	Breakeven
SW-0020	0.243	6.0	-0.016	4.84
SW-0021	-0.236	6.0	-1.560	-0.85
SW-0022	-0.314	6.0	-0.407	-15.16
SW-0023	-0.059	1.5	-0.957	-0.07
SW-0024	-0.119	1.5	-2.198	-0.06
SW-0025	-0.010	1.5	-0.410	-0.01
SW-0026	0.068	1.5	-0.412	0.21
SW-0027	-0.305	1.5	-3.416	-0.18
SW-0028	-0.184	1.5	-0.422	-0.94
SW-0036	0.191	10.0	-0.410	2.25
SW-0039	0.710	10.0	0.403	29.00
SW-0040	-0.710	10.0	-0.939	-29.00
SW-0041	0.814	10.0	0.468	30.68

C Variant distributions

Table 9: Annualised Sharpe distribution across variants within each sweep. Pos. is the share of variants with positive Sharpe.

ID	Min	P05	P25	Median	P75	P95	Max	Pos.
SW-0002	-3.33	-1.94	-1.19	-0.55	0.03	0.35	0.64	27.0%
SW-0005	-0.76	-0.55	-0.26	0.06	0.33	0.70	0.96	54.4%
SW-0006	-1.05	-0.49	-0.23	0.10	0.38	0.75	1.13	54.6%
SW-0007	-2.46	-0.91	-0.33	-0.07	0.23	0.63	1.26	42.0%
SW-0008	-3.72	-1.76	-0.49	-0.13	0.26	0.63	1.11	41.5%
SW-0009	-2.59	-1.29	-0.59	-0.13	0.21	0.70	0.96	38.1%
SW-0010	-5.08	-4.40	-3.63	-1.46	-0.00	0.36	0.65	24.8%
SW-0011	-2.74	-1.49	-0.76	-0.54	-0.28	0.22	0.81	13.0%
SW-0012	-2.36	-1.93	-1.06	-0.56	-0.09	0.48	1.09	22.7%
SW-0013	-4.88	-2.33	-1.22	-0.56	-0.04	0.50	1.10	22.5%
SW-0014	-6.78	-2.93	-0.80	-0.36	0.05	0.46	1.10	27.9%
SW-0015	-5.17	-3.19	-1.73	-1.02	-0.36	0.38	0.88	15.4%
SW-0016	-17.61	-10.91	-7.94	-2.85	-0.28	0.38	0.79	18.2%
SW-0017	-2.38	-1.98	-1.34	-0.63	-0.04	0.28	0.54	19.4%
SW-0018	-4.42	-3.27	-2.33	-1.07	-0.44	0.17	0.35	7.7%
SW-0019	-2.70	-1.40	-0.81	-0.35	-0.02	0.29	0.64	23.5%
SW-0020	-2.47	-0.78	-0.34	-0.02	0.37	0.92	1.69	48.1%
SW-0021	-5.89	-4.34	-2.75	-1.56	-0.42	0.17	0.55	10.7%
SW-0022	-2.77	-1.38	-0.69	-0.41	-0.11	0.17	0.41	13.0%
SW-0023	-5.82	-4.28	-2.35	-0.96	-0.34	0.35	0.65	12.0%
SW-0024	-8.53	-6.67	-4.21	-2.20	-0.40	0.34	0.87	12.6%
SW-0025	-4.85	-2.02	-0.83	-0.41	0.00	0.39	0.85	25.3%
SW-0026	-4.47	-2.01	-0.95	-0.41	-0.01	0.45	0.80	22.4%
SW-0027	-10.20	-7.67	-5.24	-3.42	-0.51	0.17	0.68	8.4%
SW-0028	-5.17	-2.11	-0.80	-0.42	-0.03	0.44	0.78	23.0%
SW-0036	-13.67	-5.08	-1.17	-0.41	-0.06	0.39	0.86	23.3%
SW-0039	-0.62	-0.23	0.09	0.40	0.72	0.88	0.97	82.2%
SW-0040	-1.92	-1.80	-1.30	-0.94	-0.62	-0.41	-0.20	0.0%
SW-0041	-0.24	-0.16	0.25	0.47	0.69	0.98	1.04	86.7%

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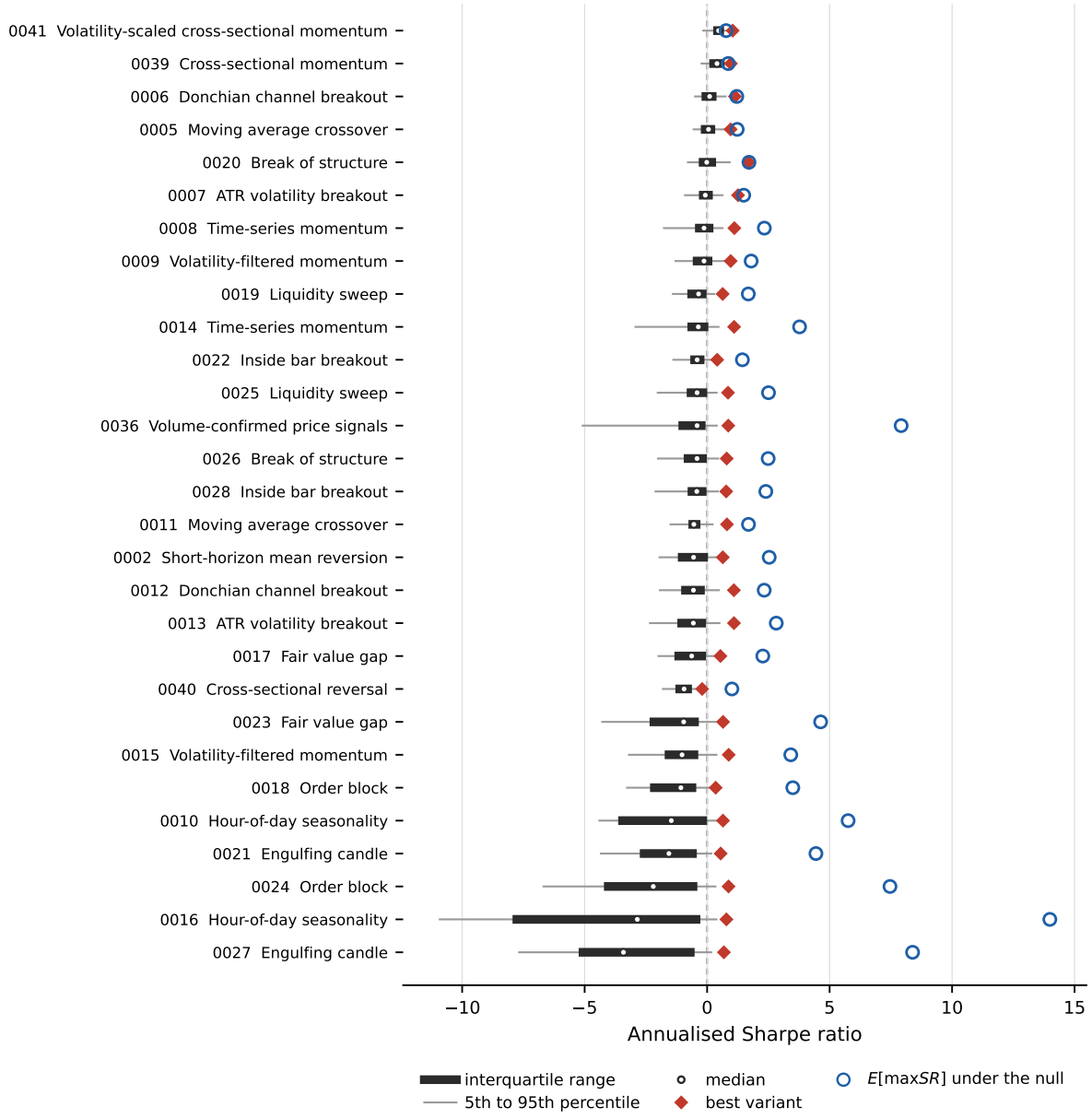


Figure 1: Distribution of annualised Sharpe ratios across variants, one row per scored sweep, ordered by median. Thick bar: interquartile range. Thin line: 5th to 95th percentile. Red diamond: best variant. Open blue circle: $E[\max SR]$ under the null. Where the circle lies to the right of the diamond, the family underperformed chance.